Cyrus Hong

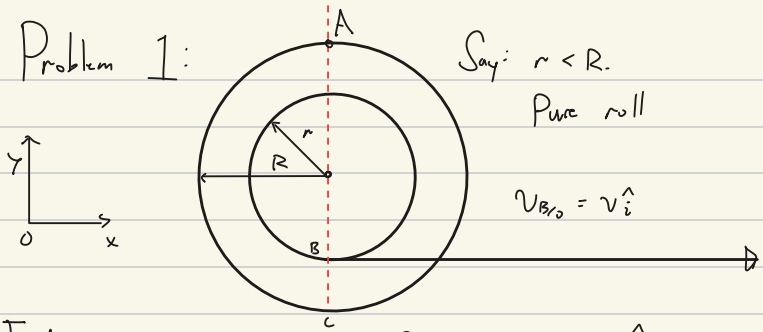
Aurorcys

MIT2.003 SC PS4



Made with Goodnotes

Problem 1:

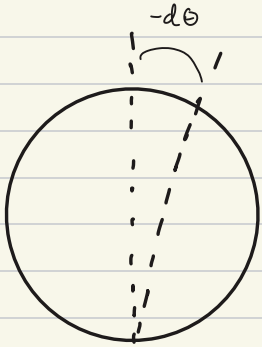


Say: $r < R$.

Pure roll

$v_{B/O} = v\hat{i}$

After some small rotation $d\theta$ (exaggerated lol)



Find instantaneous $v_{A/O}$ when B is pulled at $v\hat{i}$

Since instantaneous, C is ICR (instantaneous center of rotation)

$$v_{A/O} = v_{A/C} + v_{C/O}$$
$$= v_{A/C} + 0$$
$$= \left(\frac{\partial r_{A/C}}{\partial t}\right)_{\omega=0} + \omega_{/O} \times r_{A/C}$$
$$= 0 + \dot{\theta}\hat{k} \times 2R\hat{j}$$
$$= -2R\dot{\theta}\hat{i}$$

What is $\dot{\theta}$?

Let's use our other source of information

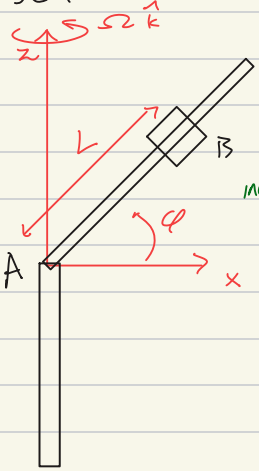
$v_{B/O} = v\hat{i}$

Finding θ:

$$v_{B/O} = v_{B/C} + v_{C/O} = \left(\frac{\partial r_{B/C}}{\partial t}\right)_{\omega=0} + \omega_{/O} \times r_{B/C} + 0$$
$$= 0 + \dot{\theta}\hat{k} \times (R-r)\hat{j}$$
$$v\hat{i} = -\dot{\theta}(R-r)\hat{i}$$
$$\dot{\theta} = \frac{-v}{R-r}$$

$$v_{A/O} = \frac{2Rv}{R-r}\hat{i}$$

Problem 2:



B. 2kg

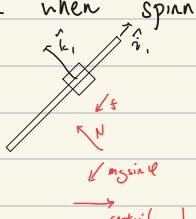
$\mu_s = 0.2$ (between spool & rod)

$L = 0.25\,m$

$\varphi = 30^\circ$

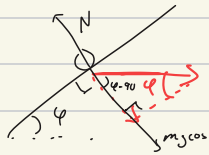
Find max (Ω) such that L does not increase.

fbd when spinning



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Use first information. $\hat{k}_1$ forces must balance out.

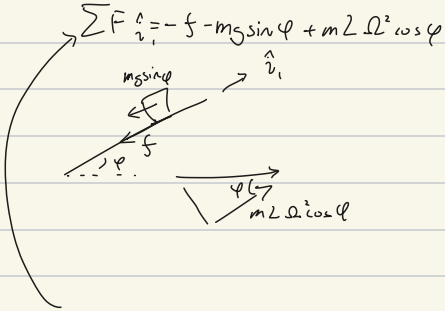


(centrifugal)

$$\sum F_{\hat{k}_1} = N - mg\cos\varphi = mL\Omega^2\sin\varphi$$

$$N = mg\cos\varphi + mL\Omega^2\sin\varphi$$

$$f = \mu N$$



$$\sum F_{\hat{i}_1} = -f - mg\sin\varphi = -mL\Omega^2\cos\varphi$$

(why negative? Think of it as fictitious force)

$$-\mu(mg\cos\varphi + mL\Omega^2\sin\varphi) - mg\sin\varphi = mL\Omega^2\cos\varphi$$

$$-mg(\mu\cos\varphi + \sin\varphi) = \Omega^2(mL\cos\varphi + \mu mL\sin\varphi)$$

$$\Omega = \sqrt{\frac{-mg(\mu\cos\varphi + \sin\varphi)}{mL\cos\varphi + \mu mL\sin\varphi}}$$

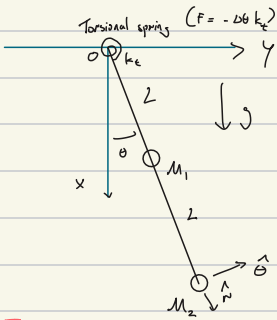
$m = 2kg$, $\varphi = 30°$, $L = 0.25m$

$$\Omega \approx 5.87 \text{ rad/s}$$

Problem 3:

a) Find five constraints that makes it only have 1 # dof

b) Find EOM considering $\frac{d}{dt}h_{/0}$, with fbd.



Finding constraints:

constraint in $\dot{x}$, $\dot{y}$, $\dot{z}$ constraint # = 3

constraint in rotation about x, y. constraint # = 2

constraint = 5 ✓

Finding EOM

$$h_{/0} = \sum_i r_{i/0} \times p_{i/0}$$

$$= r_{1/0} \times m_1 v_{1/0} + r_{2/0} \times m_2 v_{2/0}$$

$$= L\hat{r} \times m_1(\dot{\theta}\hat{k} \times L\hat{r}) + 2L\hat{r} \times m_2(\dot{\theta}\hat{k} \times 2L\hat{r})$$

$$= L^2 m_1 \dot{\theta}\hat{k} + 4L^2 m_2 \dot{\theta}\hat{k}$$

$$= (L^2 m_1 \dot{\theta} + 4L^2 m_2 \dot{\theta})\hat{k}$$

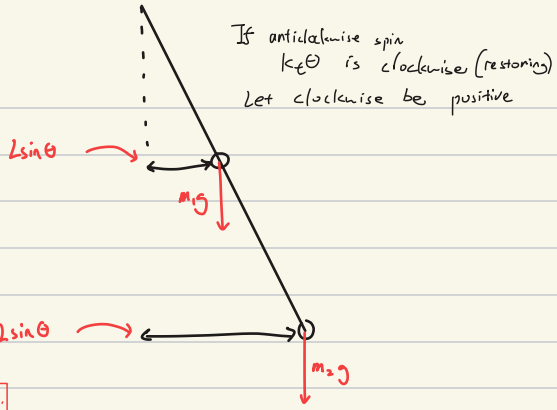
$$h_{/0} = (L^2 m_1 + 4L^2 m_2)\dot{\theta}\hat{k}$$

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$$h_{/O} = (L^2 m_1 + 4L^2 m_2)\dot{\theta}\hat{k}$$

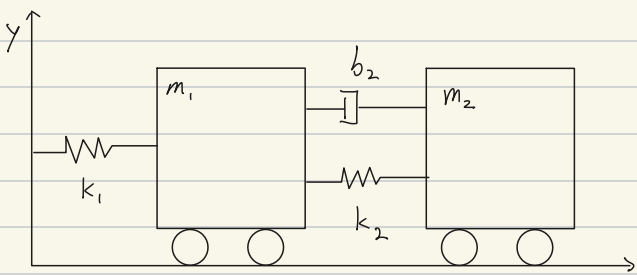
$$\therefore \frac{d}{dt} h_{/O} = \Sigma \tau$$

$$\Sigma \tau = \left[-\theta k_t - (m_1 + 2m_2) g L \sin\theta\right]\hat{k}$$



$$-\theta k_t - (m_1 + 2m_2) g L \sin\theta = (m_1 + 4m_2) L^2 \ddot{\theta}$$

Problem 4:



a) DOFs = $6N - C$

N: number of rigid bodies

C: number of constraints

This problem has 2 DOFs, find the ten C's

b) fbd, find 2 EOM

a) # of constraints, $C = 0$ init

m_1 xyz rotation, $C + = 3$

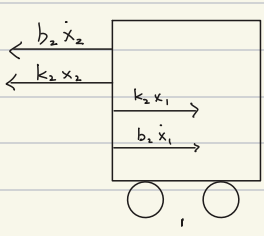
m_2 xyz rotation, $C + = 3$

m_1 & m_2 $\dot{x}$ $\dot{y}$, $C + = 2 \times 2$

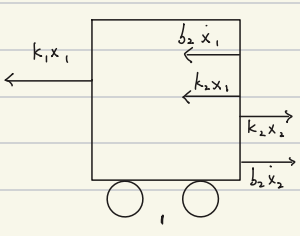
So $C = 10$ The 2 dofs are $\dot{x}$ for m_1 & m_2

b) fbds and then 2 EOM

for m_2 (assuming +ve displacement)



for m_1 (assuming +ve displacement)



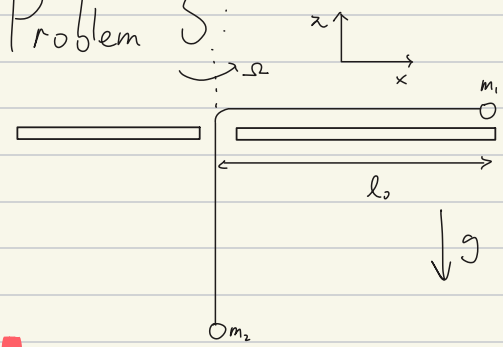
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Using $\sum F = ma$ as applied law:

$$\sum F_{on\ m_1} = k_2 x_2 + b_2 \dot{x}_2 - (k_1 x_1 + k_2 x_1 + b_2 \dot{x}_1) = m_1 a_{1/0}$$

$$\sum F_{on\ m_2} = k_2 x_1 + b_2 \dot{x}_1 - (k_3 x_2 + b_3 \dot{x}_2) = m_2 a_{2/0}$$

Problem 5:



frictionless

a) find # dofs

b) find EOMs

c) will, on initial conditions, m_1 move outward or inward.

a) dof equation: $6M + 3n - C$ M: # rigid bodies, n: # particles, C: # constraints

$\therefore M = 0$, $n = 2$

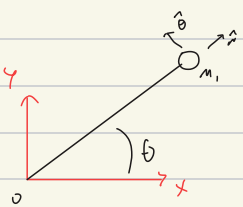
dof: $2 \times 3 - C$

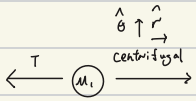
finding # of C, init $c = 0$, m_1 $\dot{z}$ $c+=1$, m_2 $\dot{y}$ $\dot{x}$ $c+=2$, m_2 $\dot{z} = m_1 \dot{r}$, $c = 4$

So # dof $= 6 - 4 = 2$, r and θ

b) finding EOMs:

fbd Top view:





Using $\sum F = ma$, and putting centrifugal on the force side

$$\sum F_r = m_1 \vec{a}_{1/0} = m_1 \ddot{r} \hat{r} = (-T + m_1 r \dot{\theta}^2)\hat{r}$$

$$T = m_1 (r\dot{\theta}^2 - \ddot{r})$$

$$\sum F_{\hat{k}} = m_2 \vec{a}_{2/0} = m_2 \ddot{z} \hat{z} = (T - m_2 g)\hat{z}$$

$$T = m_2 \ddot{z} + m_2 g$$

$$T = m_2 (\ddot{z} + g)$$

fbd side view:



Substituting expressions:

$$m_1 (r\dot{\theta}^2 - \ddot{r}) - m_2 (\ddot{z} + g) = 0$$

This is EOM number 1.

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EOM #2.

Using information. We know no torques are applied to the system. And since $\Sigma\tau = \frac{dH_{1/o}}{dt}$. There is no change to angular momentum. So it is conserved.

$H_{1/o} = \sum_i r_{i/o} \times P_{i/o}$. Since only $\hat{k}$ translation for m_2. $h_{1/o}$ for m_2 only is 0. As the positional vector and velocity vector points in the same direction.

$H_{1/o} = r_{1/o} \times m_1 v_{1/o} = r\hat{r} \times m_1 \left(v_{1/o_{w=0}} + \omega_{1/o} \times r_{1/o}\right)$. As $v_{1/o_{w=0}} = 0$.

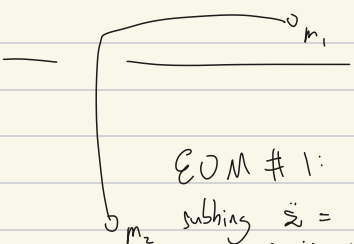
$H_{1/o} = r\hat{r} \times m_1 (\dot{\theta}\hat{k} \times r\hat{r}) = r\hat{r} \times m_1 (\dot{\theta} r \hat{\theta}) = m_1 r^2 \dot{\theta}\hat{k}$

Since $H_{1/o} = m_1 r^2 \dot{\theta}\hat{k}$. $\frac{d}{dt} H_{1/o} = (2m_1 r\dot{r}\dot{\theta} + m_1 r^2 \ddot{\theta})\hat{k} = 0$, $\quad 2\dot{r}\dot{\theta} + r^2\ddot{\theta} = 0$ ← This is the second EOM.

c) To find answer, just use EOM #1.

As the system is like this.

$\ddot{z}$ of m_2 actually is equivalent to $\ddot{r}$ of m_1. As the string is non-extending.



EOM #1: $m_1(r\dot{\theta}^2 - \ddot{r}) - m_2(\ddot{z} + g) = 0$

subbing $\ddot{z} = \ddot{r}$:

$m_1(r\dot{\theta}^2 - \ddot{r}) - m_2(\ddot{r} + g) = 0$

$-\ddot{r}(m_1 + m_2) = m_2 g - m_1 r\dot{\theta}^2$

$\ddot{r} = \frac{m_1 r\dot{\theta}^2 - m_2 g}{m_1 + m_2}$ .... (4)

If eq (4) is positive, it will begin moving outwards.

If eq (4) is negative, it will begin moving inwards.

If eq (4) is zero, it will not move (because there's no friction, so $\dot{\theta}$ is constant)

THANKS for reading.

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